<8. Analysis>

1-a: $\pi_\theta = \theta^{k-1} \cdot (1-\theta)$, $R(\pi_\theta) = k-1$

$$\log \pi_\theta = (k-1)\log\theta + \log(1-\theta)$$

$$\nabla_\theta J(\theta) = E_{\pi_\theta \sim \tau}\left(R(\tau) \cdot \nabla_\theta \log \pi \right)$$

$$= \sum_{k=1}^{\infty} \theta^{k-1}(1-\theta)\cdot(k-1) \cdot \left\{ \frac{k-1}{\theta} - \frac{1}{1-\theta} \right\}$$

$$= \sum_{k=1}^{\infty} \theta^{k-2}(1-\theta)(k-1)^2 - \theta^{k-1}(k-1)$$

$$= \sum_{n=0}^{\infty} (1-\theta)\, n^2 \theta^{n-1} - n\theta^n$$

$$= \frac{1+\theta}{(1-\theta)^2} - \frac{\theta}{(1-\theta)^2} = \frac{1}{(1-\theta)^2}$$

$$\sum \theta^n = \frac{1}{1-\theta}$$

$$\sum n\theta^n = \frac{\theta}{(1-\theta)^2}$$

$$\sum n^2\theta^{n-1} = \frac{1+\theta}{(1-\theta)^3}$$

answer

1-b

$$J(\theta) = E_{\pi_\theta}(R(\tau)) = \sum_{k=1}^{\infty} \theta^{k-1}(1-\theta)\cdot(k-1)$$

$$= \sum_{k=1}^{\infty} (k-1)\theta^{k-1} - (k-1)\theta^k = \frac{\theta-\theta^2}{(1-\theta)^2} = \frac{\theta}{1-\theta}$$

$$\nabla_\theta J(\theta) = \frac{1}{(1-\theta)^2}$$

answer

② $\overset{\nabla J(\theta)}{\overset{\shortparallel}{E[X]}} = E(R \cdot \nabla \log \pi) = \dfrac{1}{(1-\theta)^2}$

$$V[X] = E[X^2] - \{E(X)\}^2$$

$$E[X^2] = \sum_{k=1}^{\infty} \underbrace{\theta^{k-1}(1-\theta)}_{P_\theta} \cdot \underbrace{(k-1)^2}_{R^2} \underbrace{\left\{\frac{k-1}{\theta} - \frac{1}{1-\theta}\right\}^2}_{(\nabla \log \pi_\theta)^2}$$

$$= \frac{4\theta^2 + 9\theta + 1}{\theta(1-\theta)^4}$$

$$V[X] = E[X^2] - \{E(X)\}^2 = \frac{4\theta^2+9\theta+1}{\theta(1-\theta)^4} - \frac{1}{(1-\theta)^4}$$

$$= \frac{4\theta^2 + 8\theta + 1}{\theta(1-\theta)^4}$$ minimum at $\theta \approx 0.11$

③-a In this problem, $A = \hat{Q}_t$ (reward to go)

$$\nabla J(\theta) = \sum_{k=1}^{\infty} \theta^{k-1}(1-\theta)\left[\sum_{t=1}^{k}\left\{\nabla_\theta \log \pi_\theta(t) \cdot \sum_{t'=t}^{k} r_{t'}\right\}\right]$$

$$= \sum_{k=1}^{\infty} \theta^{k-1}(1-\theta) \sum_{t=1}^{k}\left\{(\text{이번 } \pi) \cdot \underbrace{(\text{이번 보상})}_{\text{reward}}\right\}$$

$$= \sum_{k=1}^{\infty} \theta^{k-1}(1-\theta) \sum_{t=1}^{k} \frac{1}{\theta}(k-t) = \sum_{k=1}^{\infty} \theta^{k-1}(1-\theta)\frac{k(k-1)}{2\theta} = \sum_{k=1}^{\infty} \frac{1-\theta}{2}(k^2-k)\theta^{k-2}$$

$$= \frac{1}{(1-\theta)^2}$$ → The answer is exactly same as the Q1.

→ unbiased.

3-b

$$E[X] = E\left[\sum_{t=1}^{\infty} \log \pi(a_t|s_t) \cdot \hat{Q}_t\right]$$

$$V[X] = E[X^2] - \{E[X]^2\}$$

$$E_{\pi}[X^2] = \sum_{k=1}^{\infty} \theta^{k-1}(1-\theta) \frac{k^2(k-1)^2}{4\theta^2} = \frac{\theta^2+4\theta+1}{\theta(1-\theta)^4}$$

$$E[X]^2 = \frac{1}{(1-\theta)^4}$$

$$V[X] = \frac{\theta^2+3\theta+1}{\theta(1-\theta)^4}$$

$$V_{orig} - V_{rtg} = \frac{3\theta+5}{(1-\theta)^4} > 0$$

$\Rightarrow$ reward-to-go's variance is always lower than Orig

4-a $\pi(a_H|s_t) = \theta^{H-t}$ $\pi_\theta(\tau^*) = \theta^{H-1}$

$$\nabla_\theta J(\theta) = E_{\tau\sim p_\theta'(\tau)}\left(\frac{\cancel{p_\theta(\tau)}}{p_\theta'(\tau)} \nabla_\theta \log \pi_\theta R(\tau)\right)$$

$$= E_{\tau\sim p_\theta'(\tau)}\left[\frac{\theta^{H-1}}{\theta'^{H-1}} \frac{H-1}{\theta} 1\right]$$

$$= (\theta')^{H-1} \times \frac{\theta^{H-1}}{(\theta')^{H-1}} \times \frac{H-1}{\theta} \times 1 = \underline{\underline{\theta^{H-2}(H-1)}}$$

4-b $E[x] = \theta^{H-2}(H-1)$

$$V[x] = E[x^2] - \{E[x]\}^2$$

$$E[x^2] = (\theta')^{H-1} \times \left(\frac{\theta^{H-1}}{(\theta')^{H-1}} \times \frac{H-1}{\theta} \times 1\right)^2 = \frac{\theta^{2H-4}}{(\theta')^{H-1}}(H-1)^2$$

$$\{E[x]\}^2 = \theta^{2H-4}(H-1)^2$$

$$V[x] = \theta^{2H-4}(H-1)^2\left\{\frac{1}{(\theta')^{H-1}} - 1\right\}$$

$$\left(\frac{\theta^2}{\theta'}\right)^H$$